

Banglish Hate Speech Detection: A Benchmark Dataset & Comparative Framework for Binary & Multi-class Classification

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Abstract— The widespread use of social media in Bangladesh has led to a growing volume of user generated content written in Banglish-Bangla expressed in English alphabets. Despite its prevalence, the lack of dedicated linguistic resources and annotated corpora has hindered effective hate speech detection in this low-resource language. To address this gap, we present a Banglish hate speech detection framework built on a manually annotated dataset comprising 31,834 samples for binary classification (hate vs. not hate) and 13,077 samples for multi-class categorization across six categories: Personal, Religious, Sexual, Geopolitical, Crime, and Political. Following systematic preprocessing and noise removal, six models SVM, Bi-LSTM multilingual BERT (mBERT), XLM-RoBERTa, BanglishBERT, and DistilBERT were trained and evaluated. Experimental results show that transformer based models consistently outperform traditional approaches, with mBERT achieving the highest accuracy of 91.2% for binary hate detection and 71.4% for multi-class classification. The proposed framework provides a foundational resource for Banglish hate speech research and paves the way for future advancements in automated social media moderation.

Keywords: *Banglish, Hate Speech Detection, Transformer Models, mBERT, Low-Resource Language.*

I. INTRODUCTION

Online platforms such as Facebook, YouTube, and X (formerly Twitter) have become major spaces for public discourse, enabling large-scale communication across diverse linguistic communities. However, the same scale that facilitates connectivity also amplifies hate speech and harassment, creating an urgent need for effective automated moderation systems. In multilingual regions like Bangladesh, this challenge is further complicated by the widespread use of Banglish, where Bangla is written using the Roman script or English alphabet and frequently mixed with English words, phrases, and syntax. Unlike standard Bangla written in native script, Banglish exhibits high lexical variation, inconsistent spelling, phonetic transliteration, and creative obfuscation, making it particularly difficult for conventional NLP systems to process reliably [1], [2]. Recent surveys indicate that while

fine-tuned transformer models often achieve F1 scores exceeding 90% on large, curated datasets written in native scripts, their performance degrades significantly on Romanized and code-mixed inputs without targeted normalization, domain adaptation, or script-aware training strategies. Moreover, many publicly available hate speech datasets are biased toward a single platform, focus primarily on high-resource or native-script languages, and provide limited transparency regarding data preprocessing. These limitations restrict cross-study comparability and reduce the practical applicability of such models in real-world deployment scenarios involving noisy, user-generated Banglish content [1].

Early research on Bengali hate speech demonstrated that classical and recurrent models can provide strong baselines. For instance, an SVM achieved 87.5% accuracy on a dataset of approximately 30,000 comments [3]. However, such models rely heavily on sparse lexical features, making them brittle to unseen romanizations, spelling variations, and intentional obfuscation issues that are central to Banglish text. Furthermore, this benchmark did not explicitly target Banglish. Attention-based RNN approaches later reported around 77% accuracy, highlighting the importance of contextual modeling, nevertheless, these studies were conducted on comparatively small datasets with limited code-mixing and coarse-grained labels, constraining their generalization to noisy, real-world Banglish scenarios [4]. As resources expanded, datasets such as BD-SHS, containing over 50,000 comments, established strong baselines for native-script Bangla hate speech detection, achieving F1 scores up to 91% [5]. Despite their scale and diversity, these datasets predominantly focus on Bangla written in native script and suffer from class imbalance, offering limited coverage of Banglish. Consequently, they do not directly address the robustness required for Banglish moderation. Similarly, DeepHateExplainer combined Bangla-BERT, mBERT, and XLM-R to achieve strong per-class F1 scores 91% for personal, 89% for geopolitical, and 84% for religious hate while also providing interpretability [6]. However, its

evaluation remains centered on native-script data and does not rigorously examine fairness, calibration, or adversarial obfuscation in Banglish.

To explicitly investigate script effects, the ACL 2022 shared task released a balanced dataset of 10,000 hate and offensive posts, equally divided between native-script and Banglish [7]. On the Banglish subset, XLM-RoBERTa achieved an accuracy range of 0.858–0.865 and a Macro F1 of 0.805–0.810, while mBERT obtained an accuracy of 0.840–0.845 and Macro-F1 of 0.789. These results underscored the advantage of multilingual subword tokenization for noisy, code-mixed text. Nevertheless, the Romanized portion of the dataset remains modest in size, with limited platform diversity, which restricts external validity and generalization to unconstrained, real-world Banglish content. Joint training across scripts further improved performance MuRIL achieved an accuracy of 0.850 and Macro-F1 of 0.800 but zero and few-shot transfer between scripts remained highly sensitive to pretraining choices and normalization strategies.

Beyond text-only approaches, multimodal Bengali hate speech detection pipelines incorporating both textual and visual features have been explored, particularly for meme based toxicity. Systems combining Conv-LSTM or XLM-R text encoders with deep visual backbones such as DenseNet-161 achieved F1 scores in the range of 0.78–0.83 [8], [9]. While these results highlight the complementary nature of visual cues, existing meme datasets are relatively small, computationally expensive to process, and provide shallow coverage of Banglish, limiting conclusions about scalable, production ready multimodal moderation systems.

Focusing specifically on transliterated Banglish at scale, the BanTH dataset introduced 37,350 Banglish comments with both binary and multi-label annotations [10]. TB-mBERT achieved the best binary performance with a Macro-F1 of 77.36 and accuracy of 82.57, while TB-BERT performed best on multi-label classification with a Macro-F1 of 30.17 and Hamming Loss of 7.18; XLM-R also demonstrated competitive results. Despite these advances, multi-label performance remains relatively low, reflecting the difficulty of detecting overlapping and implicit hate categories such as political-religious intersections. Additionally, while BanTH is large for Banglish text, it does not comprehensively address multimodality, cross-platform distribution shifts, or fine-grained category balance. Dialect-focused efforts, such as work on Chittagonian, suggest that lexicon-driven features can aid low-resource scenarios, but their portability to broader Banglish and resilience to creative obfuscation remain limited without complementary learned representations [11].

From the existing studies, we identify several unresolved research gaps: (1) Romanized Bangla (Banglish), characterized by inconsistent transliteration and frequent code-mixing, remains underrepresented in large-scale, real-world hate speech datasets (2) existing Banglish resources lack a unified framework supporting both binary and fine grained multi-class classification under consistent preprocessing and evaluation settings; and (3) comprehensive benchmarking of classical and multilingual transformer models on Banglish text reflecting real-world noise and deployment realism is still absent.

The key contributions of this paper are as follows:

1. We introduce a newly curated and manually annotated Banglish hate speech dataset collected from real-world

user-generated content on Facebook and YouTube. The dataset contains 13,077 comments labeled across six fine-grained categories: Personal, Geopolitical, Political, Sexual, Religious, and Crime. Annotation was performed with explicit category definitions to ensure semantic consistency in Banglish text.

2. We developed an expanded binary classification dataset comprising over 31,000 Banglish comments by integrating additional non-hate samples and remapping all annotated instances into hate and not hate classes. This enables robust binary hate speech detection suitable for real world moderation pipelines.
3. A detailed comparative evaluation is conducted using six classical and transformer-based models SVM, Bi-LSTM, mBERT, XLM-RoBERTa, BanglishBERT, and DistilBERT on both the multi-class and binary versions of the dataset under consistent preprocessing and evaluation protocols.
4. Through extensive experimentation, we analyze the strengths and limitations of multilingual pretrained models on Banglish data and identify mBERT as the top-performing model for both binary and multi-class Banglish hate speech detection, providing actionable insights for future research and deployment.

II. METHODOLOGY

This section describes the complete implementation process of the proposed Banglish hate speech detection framework.

A. Dataset Description & Annotation

A Banglish hate speech dataset was developed by collecting publicly accessible user comments from Facebook and YouTube using custom web-scraping scripts. The corpus captures naturally occurring Banglish communication. On social media, it's common to see Banglish: a casual mix of English and Bangla words, both written in English letters. To ensure ethical compliance, only publicly available content was included.

A core team of four native Bangla speakers, trained in social media analysis and Banglish features, manually annotated the dataset using detailed guidelines that defined hate speech across six categories: Personal, Religious, Sexual, Political, Geopolitical, and Crime. The process involved two phases: (i) binary classification (“hate”/“not hate”), and (ii) categorization of hate instances. To ensure quality, two independent validators reviewed all annotations, with ambiguous cases removed. Only labels agreed upon by both annotators and validators were retained, yielding high reliability (Fleiss’ $\kappa = 0.88$). The final consensus labels form the dataset’s gold standard..

kintu eta bharoter shorojontro ar vondami	Geopolitical
ei chor gula ke guli kora uchit	Crime
madarchod der kete fela uchit	Crime
chagoler baccha siyam	Personal
ei shala tora ki bhabe eisob byasha potita meyeder niye toks	Sexual
bnp netader ke chudi	Political
tor bap pagol naki	Personal
shala neta gula desh bikri kore	Political
tora sob hindu kuttar baccha	Religious

Fig. 1. Sample of hate speech classification

esob bonder aini bebosta neya huk	not hate
khanki magi tor bhoda age phatano dorkar	hate
kiyamot khub kache	not hate
madari shala bessha	hate
salare fasi hote hobe na hole kutta diye choda dite hobe	hate

Fig. 2. Sample of combined binary dataset

Fig. 1 & 2 show Sample annotated Banglish hate speech comments from the proposed dataset.

Two classification schemes were designed to support both binary and fine-grained analysis. In the binary setting, the dataset comprises 31,834 samples, with 18,757 labeled as “not hate” and 13,077 as “hate”. In the multi-class setting, the 13,077 hate instances were further distributed across the six categories: Personal (2,791), Religious (2,305), Sexual (2,247), Geopolitical (2,077), Crime (1,981), and Political (1,676). The dataset reflects the natural imbalance of hate expressions in social media, providing a realistic benchmark for model development and evaluation in low-resource language contexts.

TABLE I. DATASET DISTRIBUTION BASED ON CLASSES

Hate speech	Class	Numbers
Binary	Not hate	18757
	Hate	13077
Multi-class	Personal	2791
	Religious	2305
	Sexual	2247
	Geopolitical	2077
	Crime	1981
	Political	1676

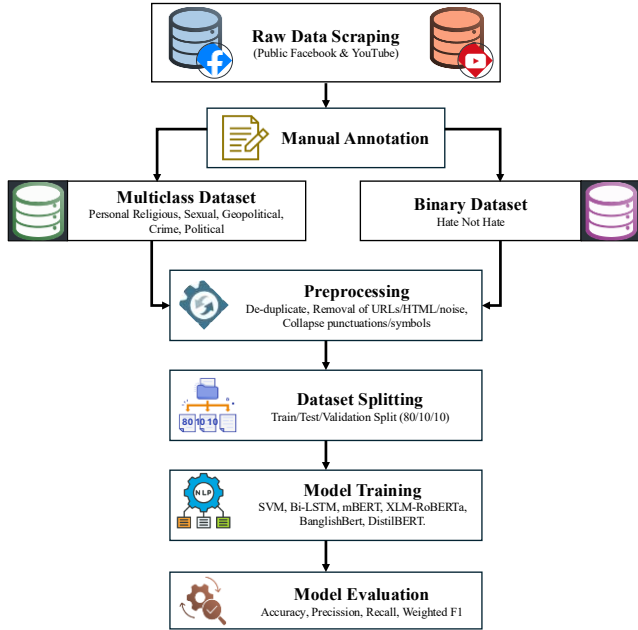


Fig. 3. Methodology of workflow

Fig. 3 shows the systematic workflow followed in this study. The dataset collection and preprocessing stages were carried out considering the linguistic variations and noise commonly present in Banglish social media text. After preparing the dataset with manual annotation and necessary cleaning procedures, six machine learning and transformer-based models were trained and validated separately for binary and multi-class hate speech detection. Finally, the performances of all models were evaluated to determine the most effective model for further analysis and benchmarking.

B. Preprocessing

Fig. 4 illustrates the preprocessing pipeline applied to the collected social media comments prior to model development. The raw dataset contained substantial noise due to informal writing styles, spelling variations, and unstructured characteristics of online discourse. To ensure data quality, duplicate entries were removed, followed by the elimination of irrelevant content such as URLs, HTML tags, excessive punctuation, random symbols, and non-textual artifacts. Although emojis may carry semantic cues, they were excluded in this study due to their sparse presence and limited influence on hate expression in Banglish text. In addition, short or context irrelevant comments were filtered out to maintain labeling reliability.

Since Banglish is a language without standardized linguistic preprocessing tools for Romanized expressions of Bangla (Banglish), stemming and lemmatization were not performed to avoid distortion of contextual meaning.

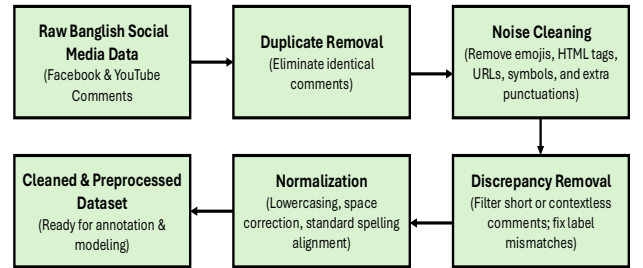


Fig. 4. Data Preprocessing pipeline.

C. Model Training

We implemented six models to evaluate both our Binary & Multi-class hate speech classification performance over our final dataset. The dataset was split into train, test, validation set. 80% of the entire dataset was taken for train set, 10% for each testing & validation. We used SVM & Bi-LSTM as a benchmark model against the powerful transformer-based models. Four transformer-based BERT models were implemented to evaluate & compare the performance across different languages & tasks to find the best model for our approach. Learning rates $1e^{-5}$, $2e^{-5}$, $3e^{-5}$ and batch sizes of 16, 32, and 64 were evaluated. After comparative analysis, the best results were achieved using a learning rate of $2e^{-5}$ and a batch size of 32. The AdamW optimizer was employed for training. Cross-Entropy Loss was used to measure the discrepancy between the model's predicted output and the true labels. Based on validation loss increase with training loss we applied early stopping to find the best epoch & eliminate the risk of overfitting. Among all models, mBERT achieved the highest accuracy & robust generalization for both the classification.

Fig. 5 illustrates the architecture of our best performing mBERT model employed in this study, consisting of three primary components: a multilingual embedding layer, a Transformer encoder stack, and a classification head. The embedding layer represents subword tokens using WordPiece tokenization and integrates positional embeddings to preserve the sequential order of text. The Transformer encoder comprises multiple stacked self-attention layers that capture contextual dependencies across Banglish text by leveraging multilingual semantic representations learned from Wikipedia corpora across 104 languages. The final

classification head is attached to the [CLS] token representation to predict hate speech categories for both the binary and multi-class classification tasks.

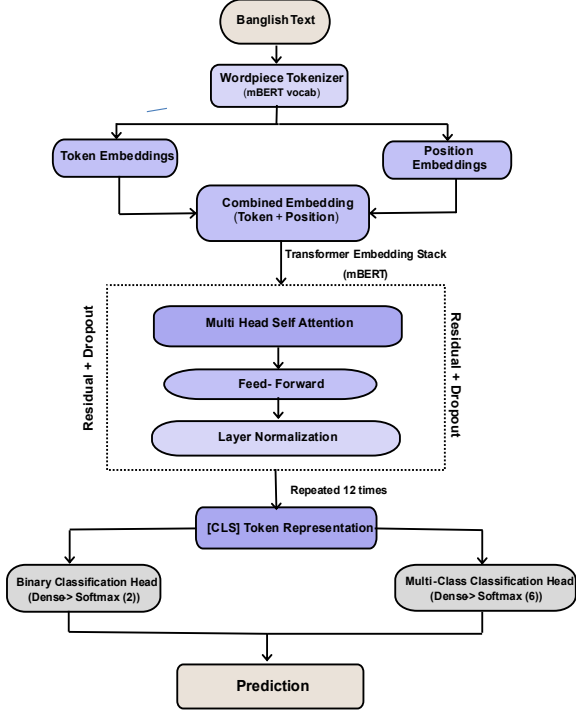


Fig. 5. Architecture of mBERT model.

D. Model Evaluation

The performance of all trained models was assessed using standard classification metrics, including Accuracy, Precision, Recall, and F1-score, for both binary and multi-class hate speech detection tasks. Evaluation was carried out on a stratified test split to ensure balanced representation across class labels. In addition, confusion matrices were analyzed to identify error patterns and class-wise misclassification tendencies, particularly among semantically overlapping hate categories. Among the six evaluated models, transformer-based architectures demonstrated superior performance compared to traditional machine learning methods. In particular, mBERT achieved the highest overall performance across both tasks, exhibiting improved generalization on Banglish text due to its multilingual pretraining capability.

III. MODEL RESULTS & DISCUSSION

A. Results

In our study, Accuracy, precision, recall, F1-score were measured and compared for six different NLP related models: SVM, Bi-LSTM, mBERT, XLM-RoBERTa, BanglishBERT and DistilBERT. These metrics will show the basis of how each model performs in terms of binary & multi-class hate speech detection.

TABLE II. PERFORMANCE EVALUATION OF DIFFERENT MODELS FOR BINARY HATE SPEECH DETECTION

Model	Accuracy	Weighted F1	Precision	Recall
SVM	0.87	0.87	0.87	0.87
Bi-LSTM	0.89	0.89	0.89	0.89
mBERT	0.91	0.91	0.91	0.91
XLM-RoBERTa	0.89	0.89	0.90	0.89

BanglishBERT	0.90	0.90	0.89	0.89
DistilBERT	0.88	0.88	0.89	0.88

TABLE III. PERFORMANCE EVALUATION OF MODELS FOR MULTI-CLASS HATE SPEECH CLASSIFICATION

Model	Accuracy	Weighted F1	Precision	Recall
SVM	0.68	0.69	0.70	0.68
Bi-LSTM	0.68	0.68	0.69	0.68
mBERT	0.71	0.71	0.72	0.71
XLM-RoBERTa	0.71	0.71	0.71	0.70
BanglishBERT	0.70	0.70	0.71	0.71
DistilBERT	0.70	0.70	0.71	0.71

Table II & III present a comparison of accuracy, precision, recall and weighted F1 score of different models for our binary hate speech detection tasks. Here in table II, mBERT clearly surpasses the other models in terms of all the metrics with accuracy of 91%, weighted F1 91%, recall 91% & most importantly with 1% higher precision than other transformer models. XLM-RoBERTa and DistilBERT performs almost similar & faster than the other models but after analyzing all the parameters, mBERT performs better comparing to the other models.

In Table III, we can see same type of performance parameters for multi-class hate speech detection where mBERT gives 71% accuracy, F1 score which is equal as the second best XLM-RoBERTa model but Precision & recall is found 1% higher than this model. DistilBERT & BanglishBERT's performance was quite impressive in our training & testing period but not reached similar precision like mBERT model.

TABLE IV. TRAINING VS VALIDATION VS TESTING ACCURACY FOR BINARY HATE SPEECH DETECTION

Model	Training Accuracy	Validation Accuracy	Testing Accuracy
SVM	0.895	0.869	0.869
Bi-LSTM	0.921	0.891	0.888
mBERT	0.925	0.903	0.912
XLM-RoBERTa	0.914	0.902	0.887
BanglishBERT	0.923	0.896	0.898
DistilBERT	0.906	0.899	0.879

TABLE V. TRAINING VS VALIDATION VS TESTING ACCURACY FOR MULTI-CLASS HATE SPEECH CLASSIFICATION

Model	Training Accuracy	Validation Accuracy	Testing Accuracy
SVM	0.689	0.678	0.679
Bi-LSTM	0.678	0.678	0.681
mBERT	0.717	0.709	0.714
XLM-RoBERTa	0.716	0.692	0.712
BanglishBERT	0.701	0.688	0.697
DistilBERT	0.712	0.708	0.709

Table IV & V represents the training, validation and testing accuracy of our tasks across different models. In our both tasks, the results were quite similar for all the models we have trained our dataset. Among them, mBERT achieved the highest accuracy for training (92.5%), validation (90.3%) & testing (91.2%) in binary task. XLM-RoBERTa got almost similar with just a little less testing accuracy than mBERT. Again showing the accuracy for multi-class task where the overall highest accuracy achieved by mBERT is 71.7% for training, 70.9% for validation, 71.4% for testing which is slightly outperformed XLM-RoBERTa (0.716/0.692/0.712) and DistilBERT (0.712/0.708/0.709). BanglishBERT showed

the lowest performance among all the transformer models while Bi-LSTM & SVM performed moderately good.

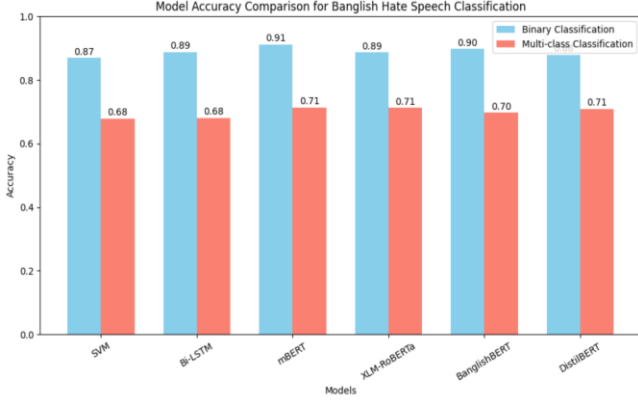


Fig. 6. Performance of both tasks for different models

Fig.6 shows the performance of both tasks for different models.

B. Performance analysis of mBERT model

Next the precision, recall, F1 score and support of our best performed mBERT model in predicting classes of Binary & multi-class hate speech detection.

Table VI shows how well binary & multi-class hate speech classification works for different classes. Precision, Recall, F1 score & Support are used as metrics to evaluate the performance of mBERT model. As most of the comments from the dataset were “not hate” Comments so the model performs well for “not hate” class with an impressive precision (92%), Recall (93%), F1 score (92%). There were low number of samples for hate comments so overall precision (90%), recall (89%) & F1 score (90%) was slightly less than not hate as we didn’t give any weight to the classes which have less samples in our dataset. For multi-class classifications, there has strict differences of the class wise performance showing in the table.

TABLE VI. CLASS-WISE PERFORMANCE EVALUATION OF mBERT MODEL FOR HATE SPEECH DETECTION

Analysis	Class	Precision	Recall	F1 Score
Binary	Not hate	0.92	0.93	0.92
	Hate	0.90	0.89	0.90
Multi-Class	Personal	0.58	0.55	0.57
	Religious	0.79	0.78	0.79
	Sexual	0.74	0.63	0.68
	Geopolitical	0.92	0.82	0.87
	Crime	0.56	0.75	0.64
	Political	0.73	0.76	0.74

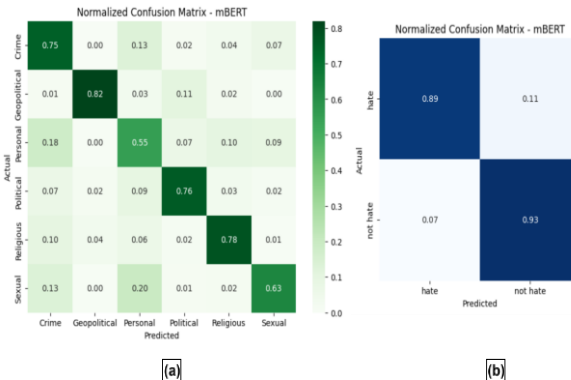


Fig. 7. Normalized confusion matrix of mBERT model. (a) multi-class classification (b) binary class classification

The confusion matrices for our best-performing model, mBERT, are shown in Fig. 7. The model’s classification performance across several binary and multi-class categories can be better understood by examining the confusion matrices. It highlights potential areas for improvement by providing a thorough analysis of the model’s predictions in comparison to the actual labels. When it comes to accurately identifying instances of “Geopolitical”, “Crime”, “Religious” & “Political” mBERT performs the best. Nonetheless, there are noteworthy cases where “Personal” and “Sexual” are confused, indicating that it may be difficult to tell these two types apart.

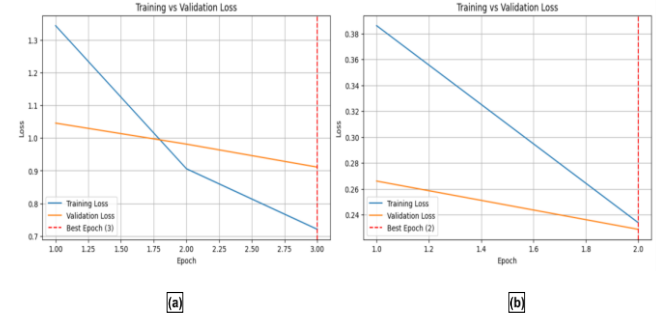


Fig. 8. Training vs Validation loss curve for mBERT model. (a) multi-class hate classification (b) Binary hate classification

Fig. 8 showing the training vs validation loss across epoch numbers to get the best epoch for binary & multi classification task. An early stopping strategy was employed, with validation loss carefully monitored to halt training once performance plateaued, thereby preventing overfitting and ensuring optimal generalization.

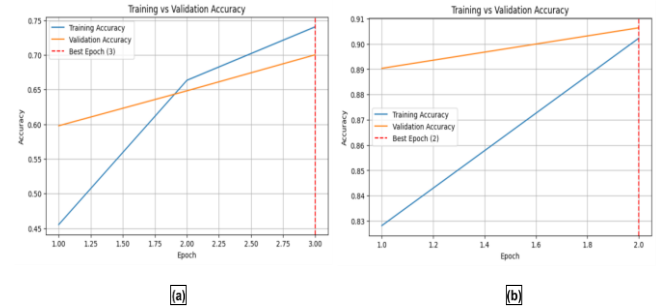


Fig. 9. Training vs Validation accuracy curve for mBERT model. (a) multi class hate classification, (b) Binary hate classification

Fig.9 illustrates the training and validation accuracy across epochs, highlighting the optimal epoch selection for both tasks.

C. Comparison of recent works

Table VIII compares the proposed mBERT model with existing approaches for hate speech detection. For external validity, we also conducted strict cross-lingual tests against other multilingual work. The proposed model consistently outperformed prior benchmarks.

TABLE VII. PERFORMANCE COMPARISON OF mBERT WITH EXISTING MODELS.

Study	Dataset Size	Model	Task	Accuracy	F1 score
N. Romim et al.[5]	50,000+	Classical + DL	Native Bangla Hate(Binary)	-----	91%
M. R. Karim et al.[6]	~30K	BanglaBERT + mBERT + XLM-R	Native Bangla Hate (Multi-claas)	-----	91% (Personal)

A. K. Das et al.[7]	10,000	XLM-R	Bangla + Banglish (Binary)	85.8–86.5%	80.5–81%
F. Haider et al.[10]	37,350	TB-mBERT	Banglish(Transliterated)	82.57%	77.36%
F. Haider et al.[10]	37,350	TB-mBERT	Banglish(Transliterated)	-----	30.17%
N. Romim. [3]	~30,000	SVM	Native Bangla (Binary)	87.5%	-----
A.K. Das et Al. [4]	7425	Attention Based Decoder	Bangla Comments on social media	77%	-----
Our work	31834	mBERT	Binary Hate (Banglish)	91.2%	91%
Our work	13077	mBERT	Multi class Hate(Banglish)	71.4%	71.1%

D. Limitations & Future Directions

Despite strong performance, the framework faces several constraints. The dataset is modest, reflecting limited Banglish content and reliance on manual annotation. The absence of standardized NLP tools (e.g., stemmers, tokenizers) restricts advanced linguistic techniques, while multilingual transformers remain under-optimized for Banglish. Dependence on a few social media platforms also introduces topical bias.

The dataset can be expanded with more diverse samples to improve coverage across topics and linguistic variations. Developing Banglish specific NLP resources would enable richer preprocessing and analysis, while fine-tuned transformer models could enhance contextual understanding and classification. Beyond text, multimodal and cross-lingual strategies may further increase robustness and generalizability in low-resource settings, advancing the applicability of hate speech detection systems.

E. Discussion & Analysis

After evaluating all the parameters such as accuracy, precision, recall and F1 score for our hate speech detection tasks, mBERT is leading the chart with best accuracy for binary hate speech detection for combined dataset with 91.2% accuracy & 71.4% for hate multiclass classification. Though DistilBERT is faster among all the models we evaluated but its accuracy & F1 score is less stable across Banglish language. On the other hand, XLM-RoBERTa offers a better F1 score & accuracy for Banglish language but it couldn't surpass the performance of mBERT in terms of overall precision, F1 score & accuracy. BanglishBERT as well didn't perform as expected for hate speech classification with only 69.7% accuracy though did better with binary hate speech detection with 89.8% accuracy. Overall, mBERT demonstrated the most balanced performance across both tasks, establishing it as the most suitable model for effective Banglish hate speech classification and future development.

IV. CONCLUSION

In this study, we developed a manually annotated Banglish hate speech dataset, comprising 13k hate comments across six

categories and an extended 31k-sample binary dataset combining hate and non-hate content from Facebook and YouTube. The dataset was preprocessed to remove noise, duplicates, and inconsistencies, ensuring clean and reliable input for model training. Six models: SVM, Bi-LSTM, mBERT, XLM-RoBERTa, BanglishBERT, and DistilBERT were trained and evaluated for both binary and multi class hate speech detection. Experimental results indicate that transformer-based architectures outperformed traditional methods, with mBERT achieving the highest overall accuracy and demonstrating strong adaptability to Banglish text. The findings establish a solid benchmark for Banglish hate speech analysis and highlight the potential of multilingual pretrained models for low resource languages. Future work will focus on expanding the dataset, developing Banglish-specific NLP tools, and implementing optimized transformer models for hate speech detection and online content moderation.

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